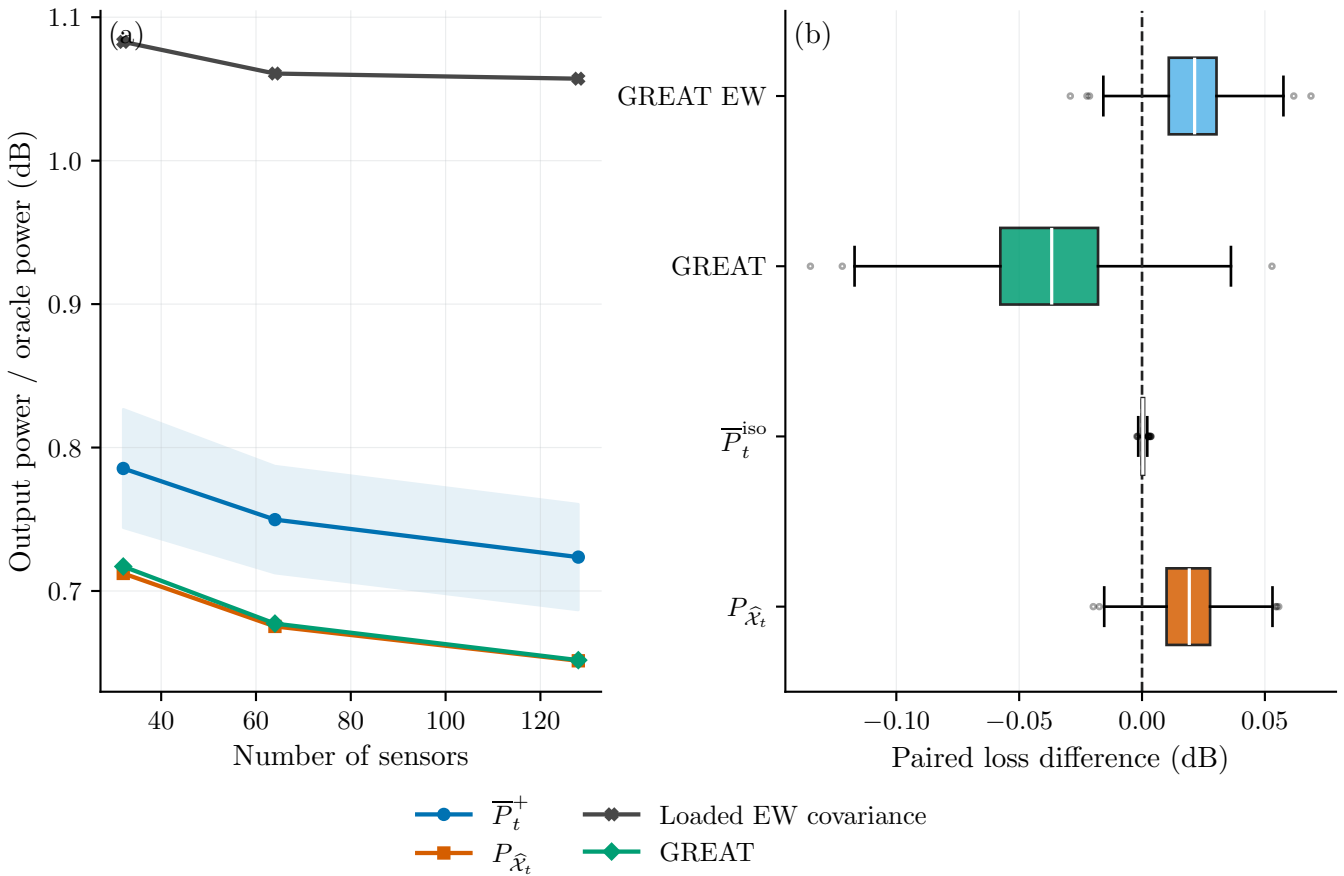
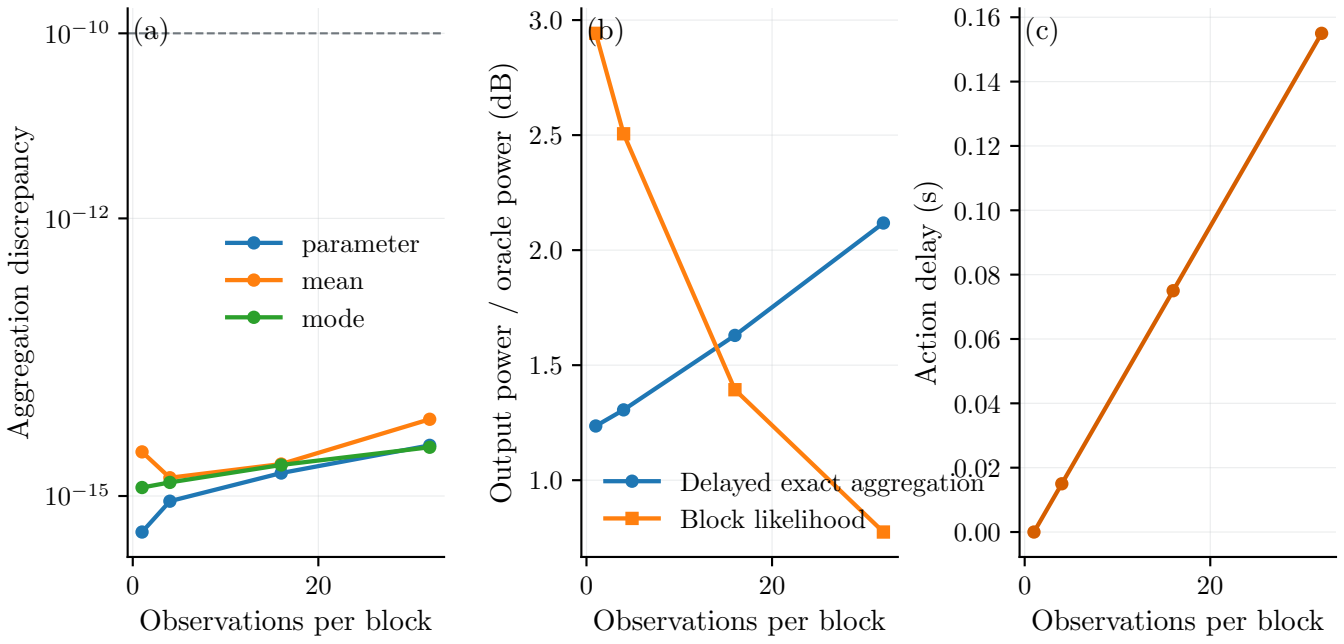
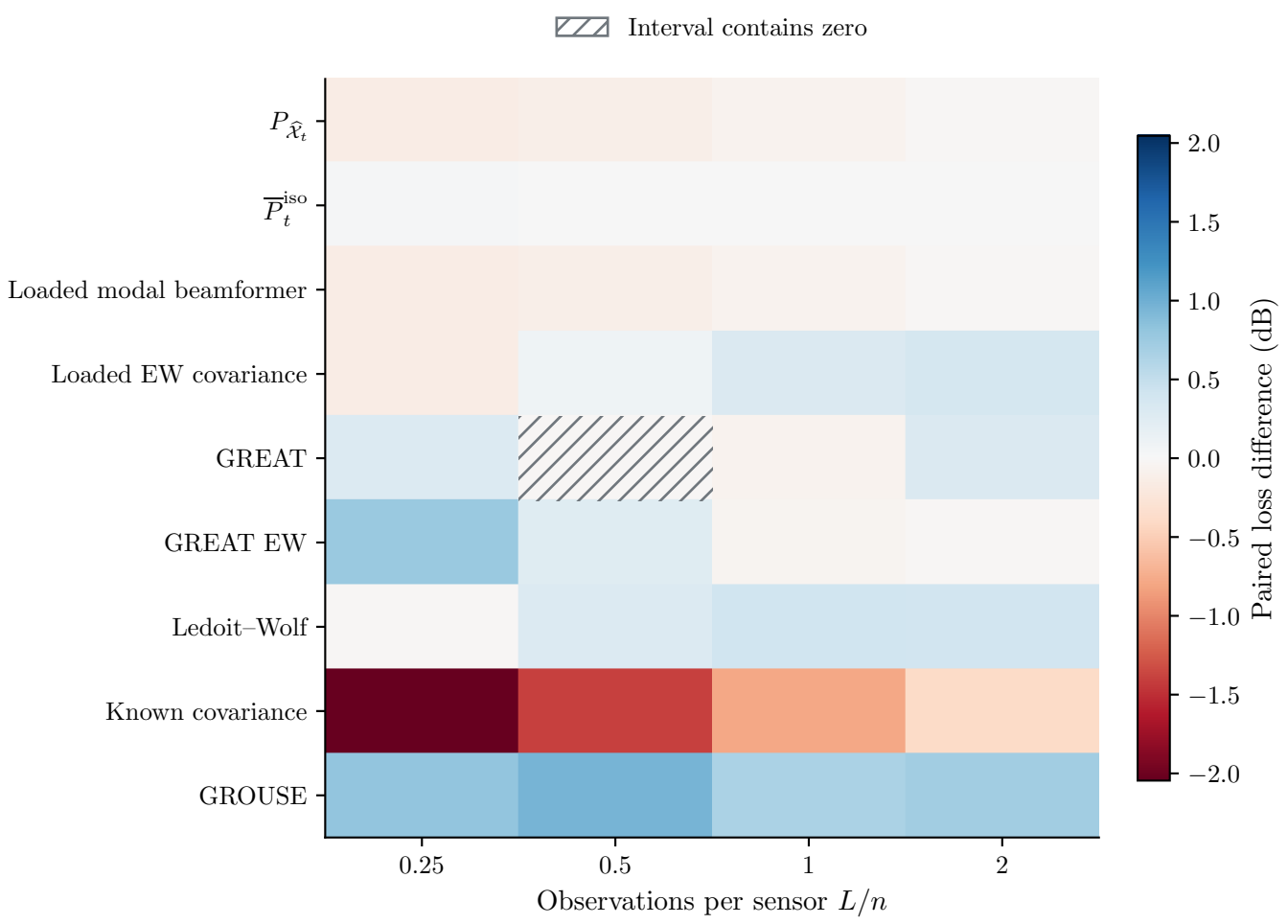
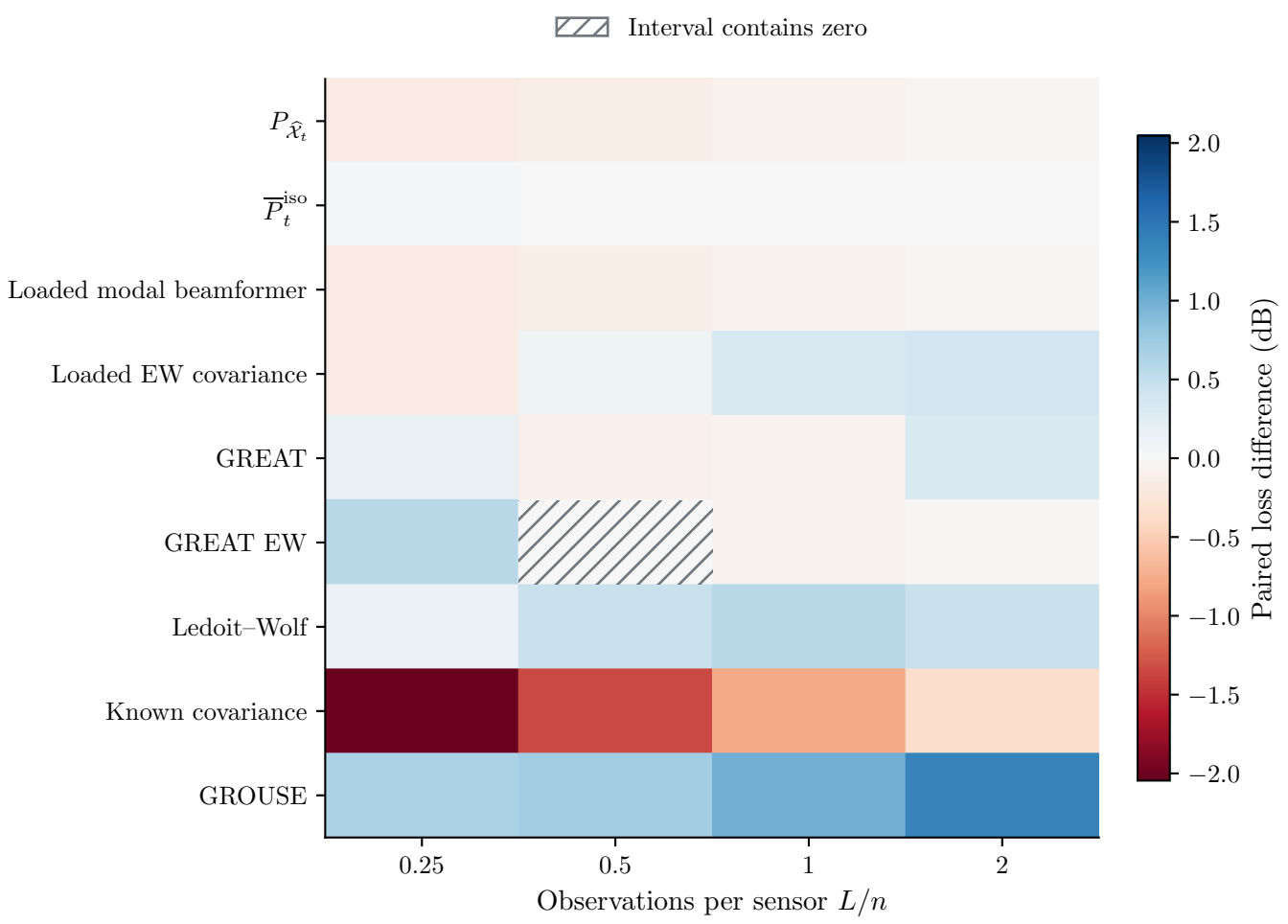
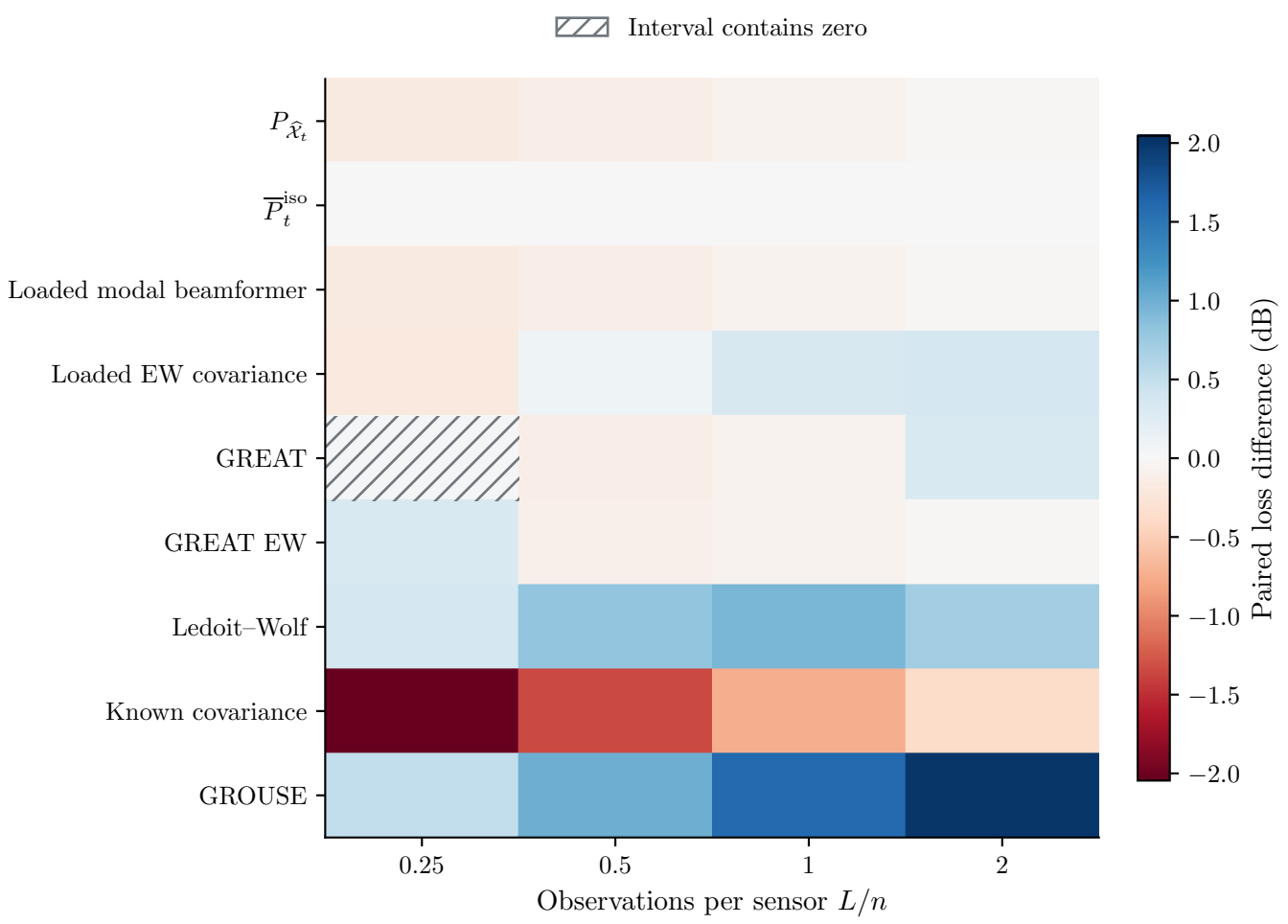
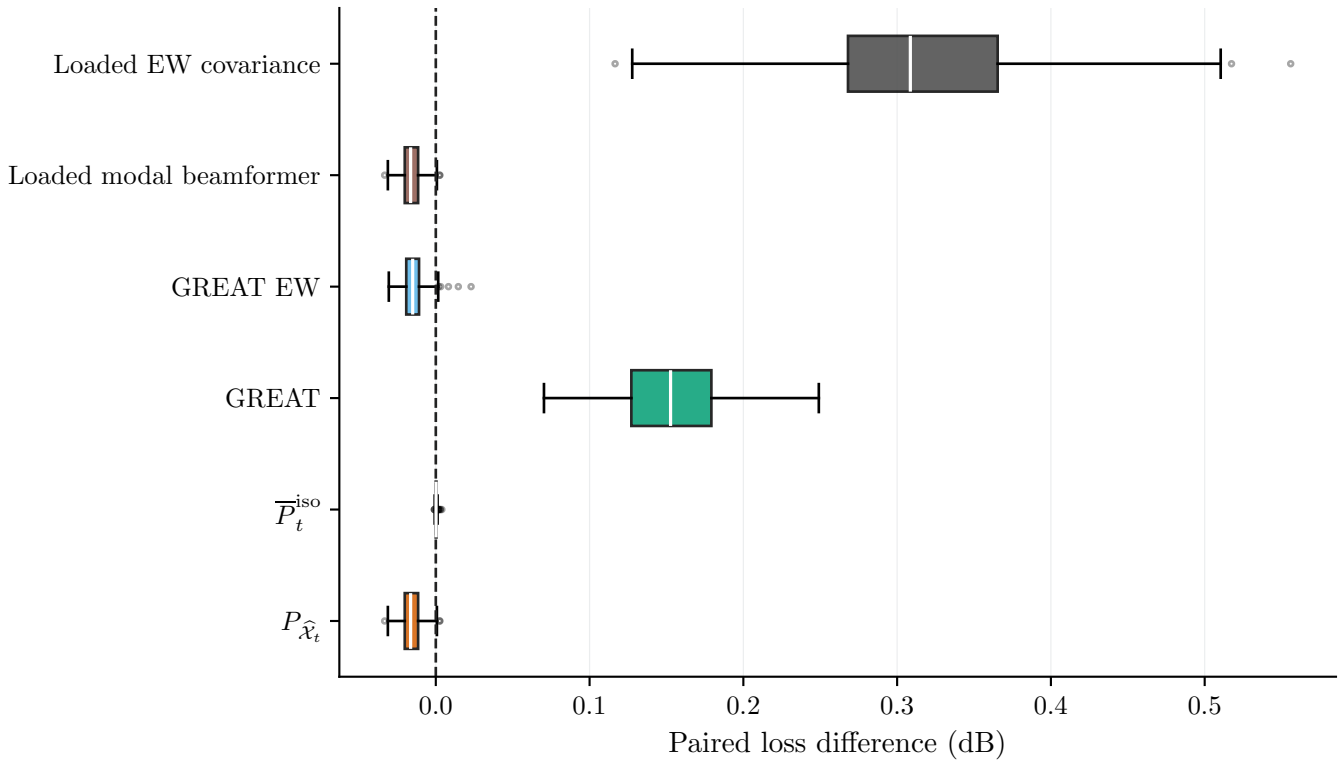

(a)
Output power / oracle power (dB)
1.1
1.0
0.9
0.8
0.7
40
60
80
100
120
Number of sensors
(b)
GREAT EW
GREAT
$\overline{P}_t^{\text{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
−0.10
−0.05
0.00
0.05
Paired loss difference (dB)
$\overline{P}_t^{+}$
$P_{\widehat{\mathcal{X}}_t}$
Loaded EW covariance
GREAT

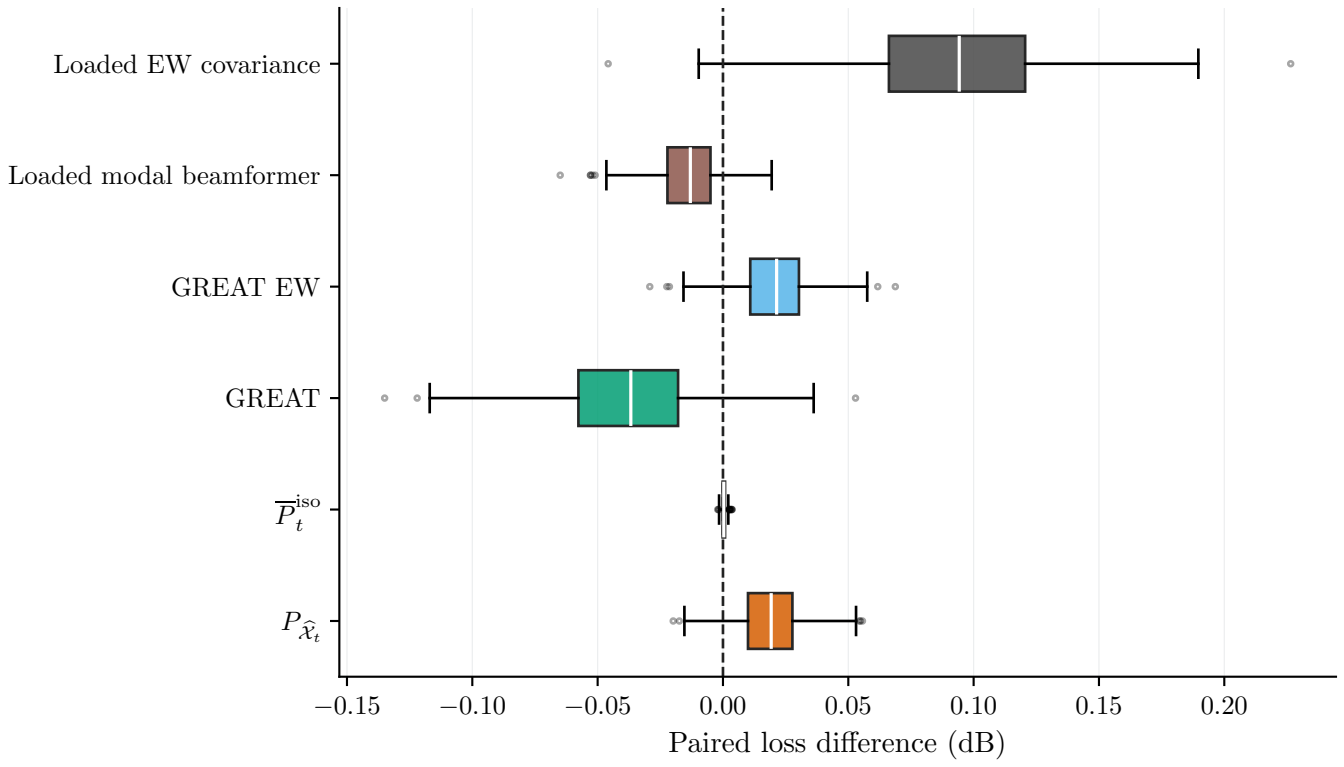

(a)
10^{-10}
10^{-12}
10^{-15}
Aggregation discrepancy
parameter
mean
mode
0
20
Observations per block
(b)
3.0
2.5
2.0
1.5
1.0
Output power / oracle power (dB)
Delayed exact aggregation
Block likelihood
0
20
Observations per block
(c)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Action delay (s)
0
20
Observations per block

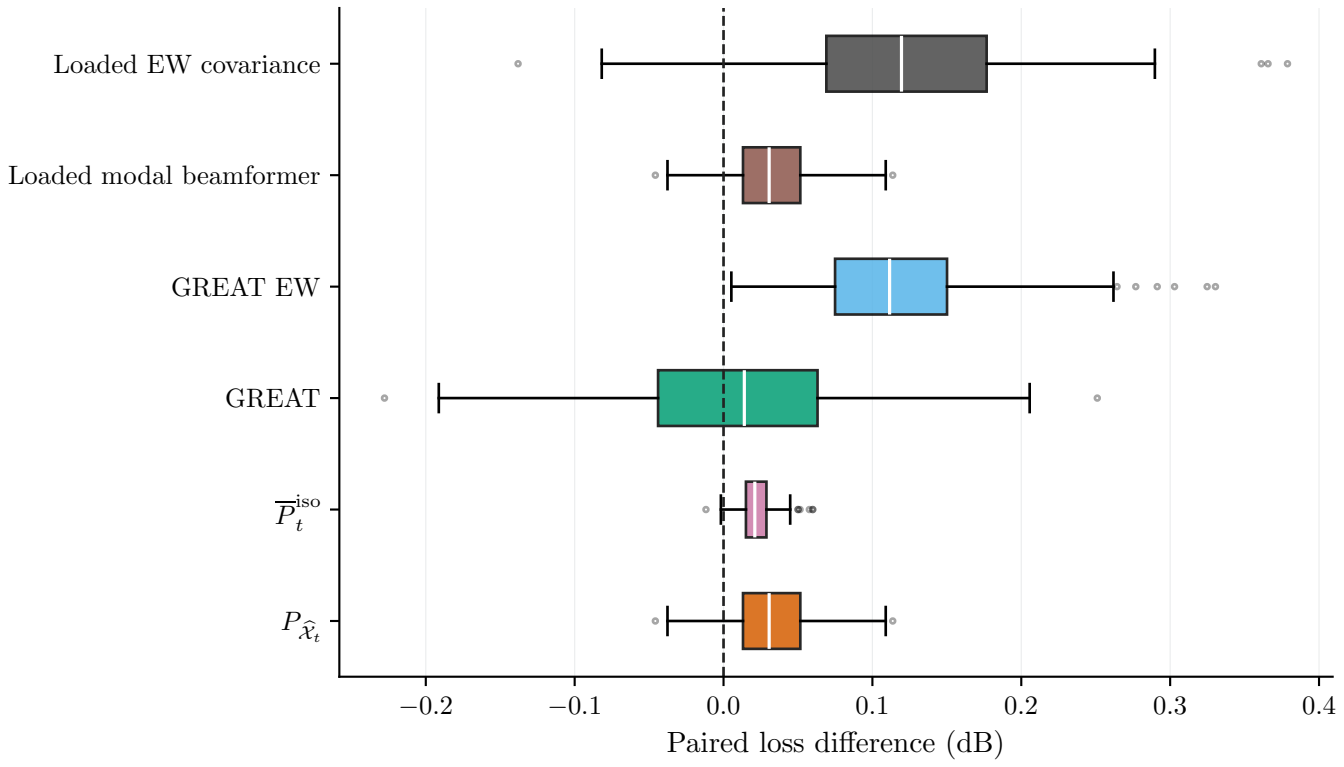

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
$\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Loaded EW covariance
GREAT
GREAT EW
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor L/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Paired loss difference (dB)

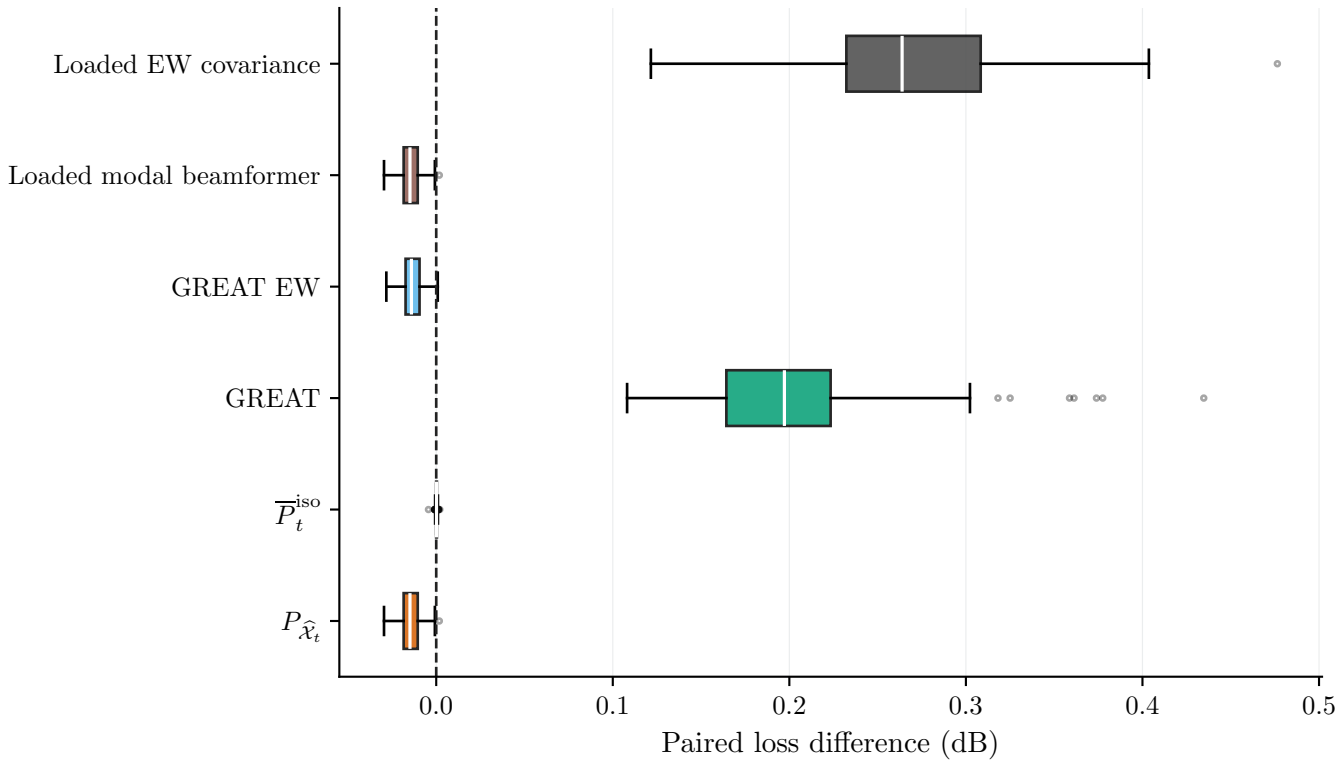

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
$\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Loaded EW covariance
GREAT
GREAT EW
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor L/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Paired loss difference (dB)

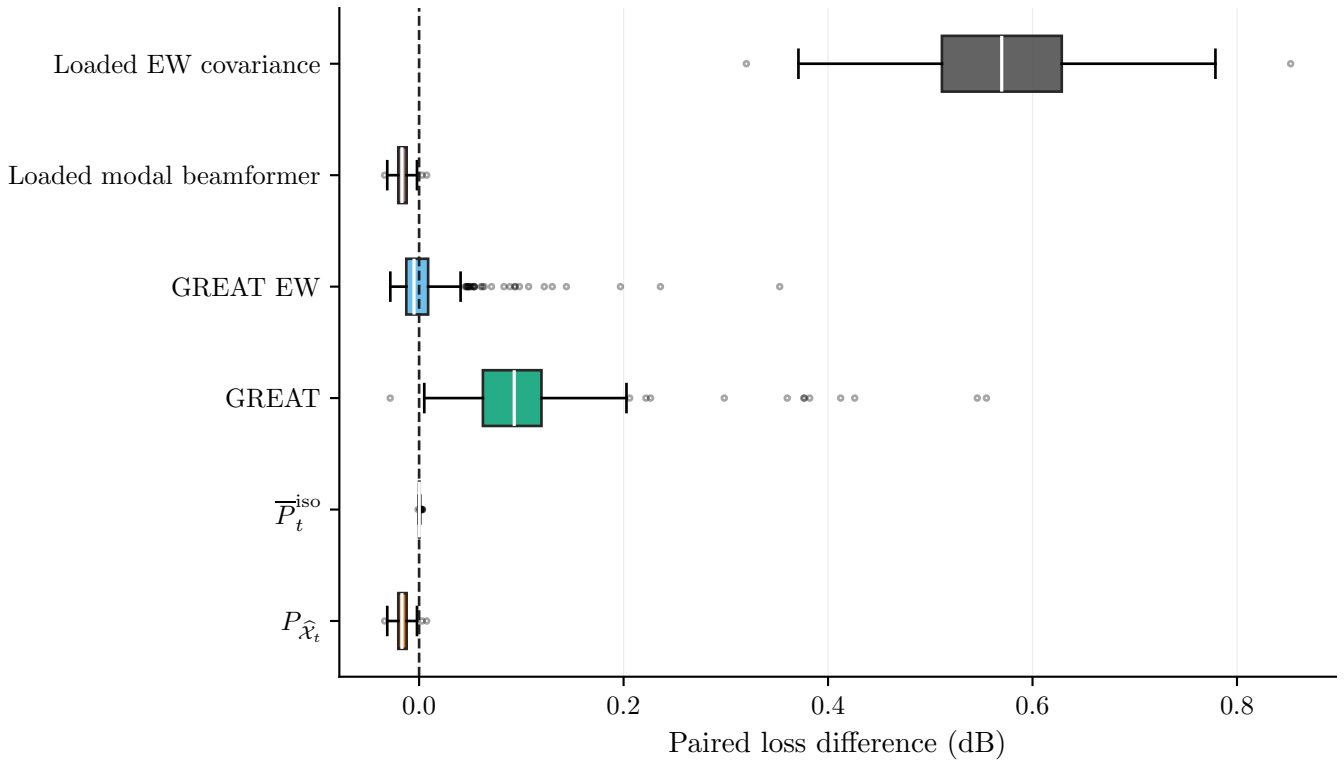

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
$\overline{P}_t^{\text{iso}}$
Loaded modal beamformer
Loaded EW covariance
GREAT
GREAT EW
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor L/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Paired loss difference (dB)

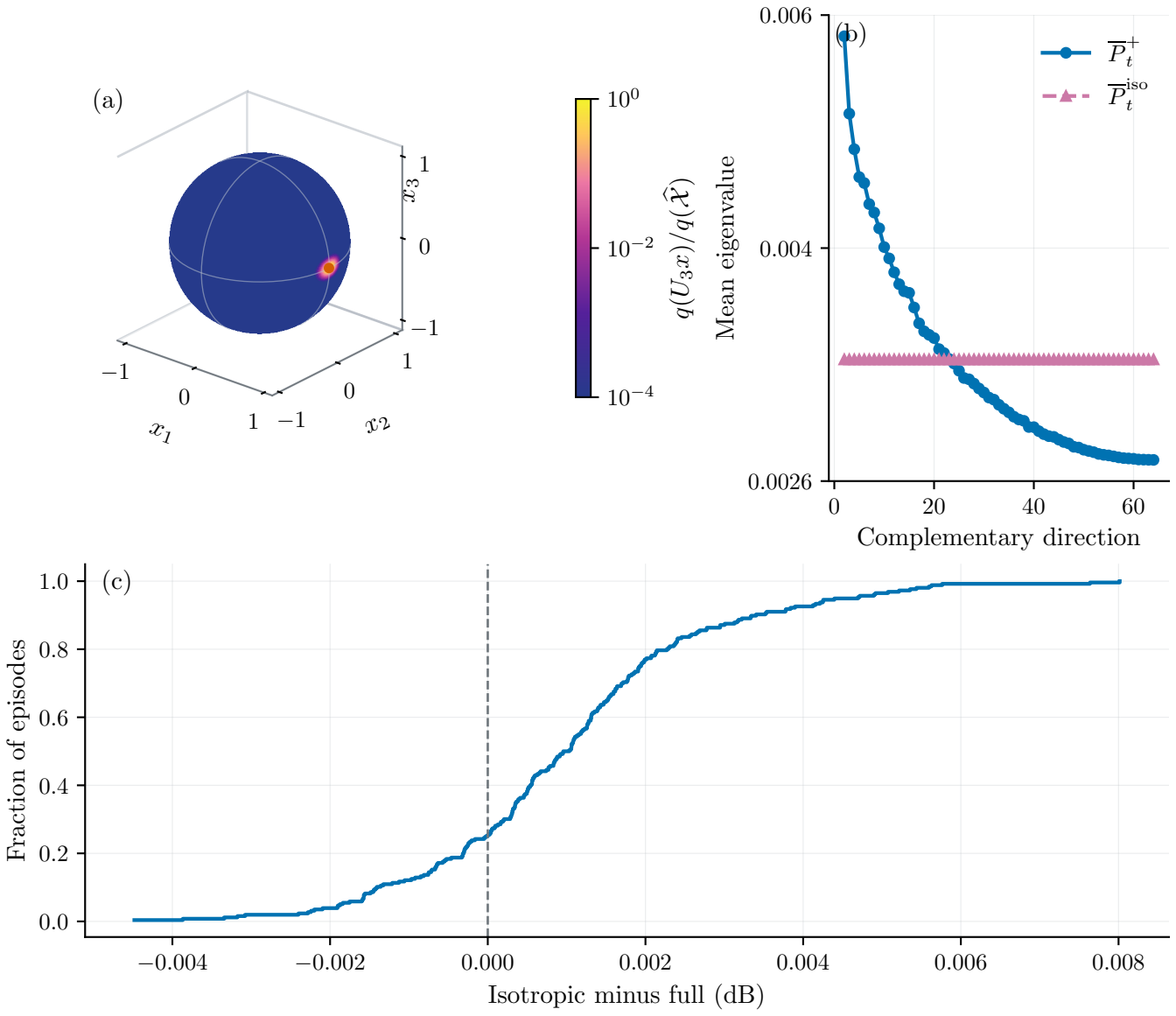

Loaded EW covariance
Loaded modal beamformer
GREAT EW
GREAT
$\overline{P}_t^{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
Paired loss difference (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
GREAT
$\overline{P}_t^{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Paired loss difference (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
GREAT
$\overline{P}_t^{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Paired loss difference (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
GREAT
$\overline{P}_t^{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
Paired loss difference (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
GREAT
$\overline{P}_t^{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.2
0.4
0.6
0.8
Paired loss difference (dB)


(a)
x_1
x_2
x_3
10^0
10^{-2}
10^{-4}
$q(U_3 x)/q(\hat{\mathcal{X}})$
(b)
$\overline{P}_t^+$
$\overline{P}_t^{\text{iso}}$
Mean eigenvalue
Complementary direction
(c)
Fraction of episodes
Isotropic minus full (dB)